Still time for action: genetic conservation of imperiled South Canadian river fishes, Arkansas Rver Shiner (*Notropis girardi*), Peppered Chub (*Macrhybopsis tetranema)* and Plains Minnow (*Hybognathus placitus*)

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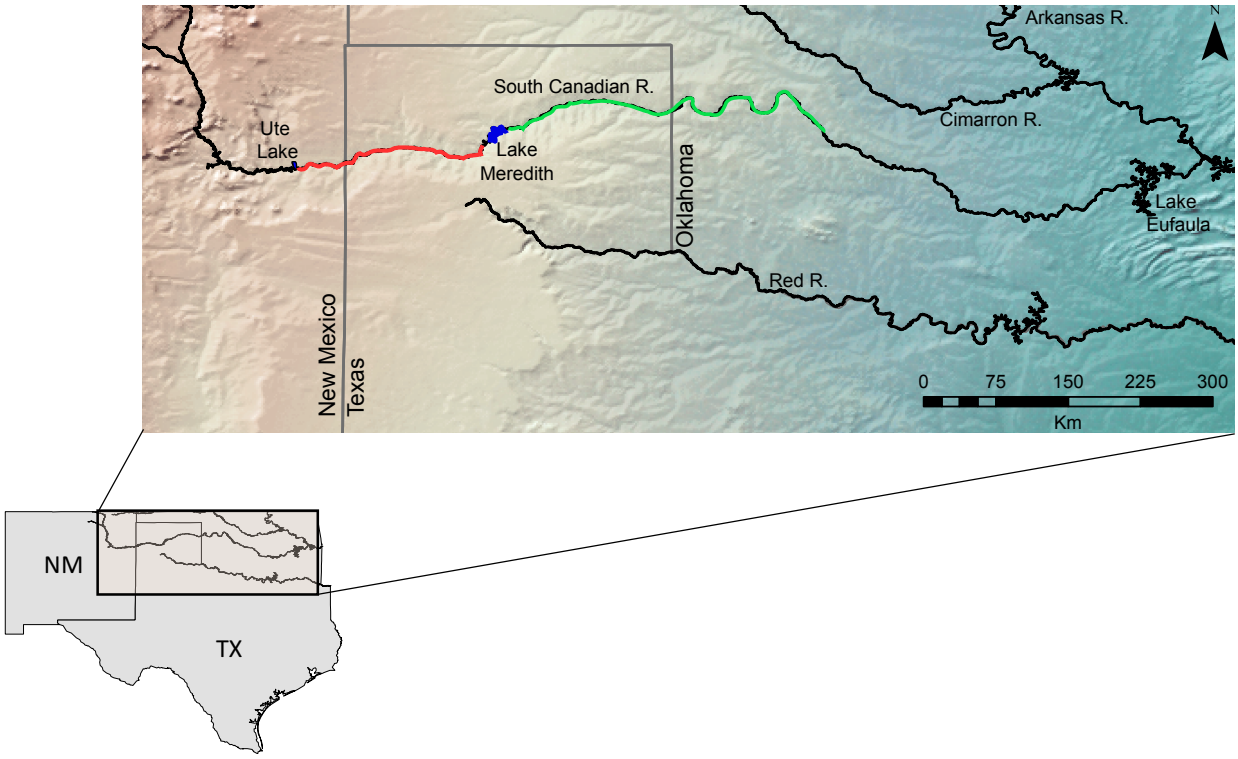
Supplementary Material

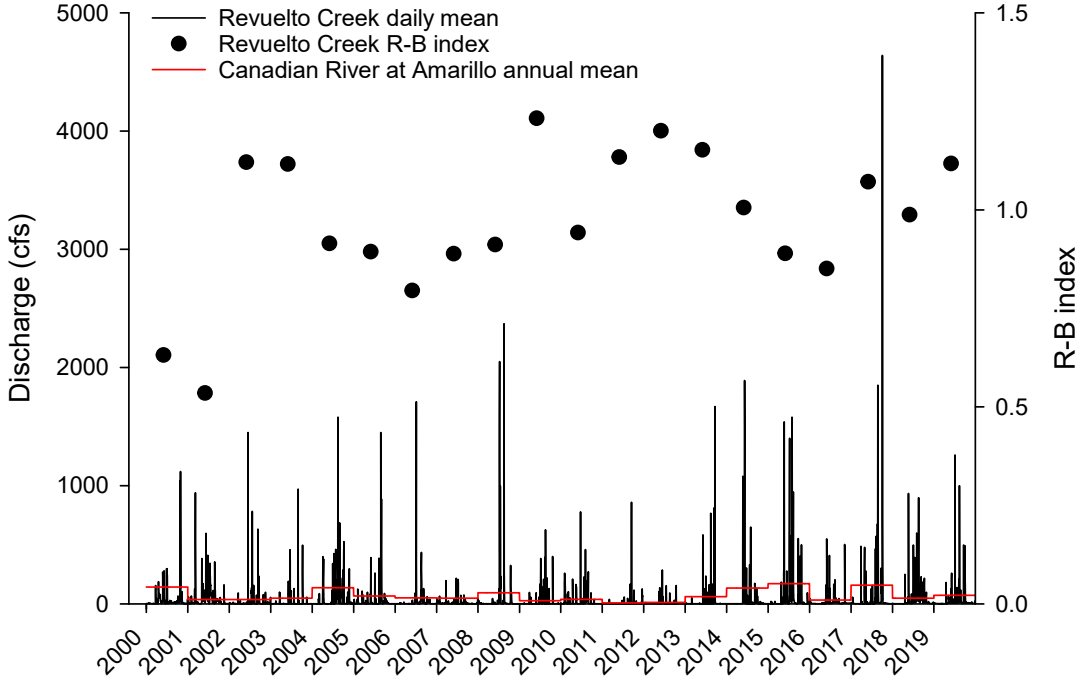
Fig. S1 Map showing locations of sampling sites for Peppered Chub (red), Arkansas River Shiner (red and green) and Plains Minnow (red and green). The location of the study area within New Mexico (NM), Texas (TX) and Oklahoma is also shown.



Ute Dam has significantly altered the natural hydrograph of the South Canadian River downstream of the dam to Lake Meredith such that there is minimal discharge from the dam and almost no variability (Dudley et al. 2012). Revuelto Creek is ephemeral and primarily flows following local rainstorms and provides important variability to the systems (Fig. S2). To illustrate the annual flow regime of the Canadian River between Ute Dam and Lake Meredith, mean annual discharge was obtained from the two USGS gages within this river fragment (Canadian River near Amarillo, TX 07227500 and Revuelto Creek near Logan, NM 07227100). An index of disturbance (R-B index, Baker et al., 2004) was calculated for Revuelto Creek. The R-B index measures day-to-day changes in discharge and is a ratio between discharge oscillation and total discharge that typically varies from 0.0 to 1.5, with higher values indicating more extreme spikes in flow.

Fig. S2 a Mean annual discharge between 2000 and 2019 at the two U.S. Geological Survey gages present in the South Canadian River above Lake Meredith, TX and discharge variability (R-B index, Baker et al., 2004) for Revuelto Creek, NM, **b** US Drought Monitor levels (percent area) for the Arkansas and Red River Basins (extreme drought [DS3] and exceptional drought [DS4]) for the period 2009-2019. https://droughtmonitor.unl.edu/

a



b.

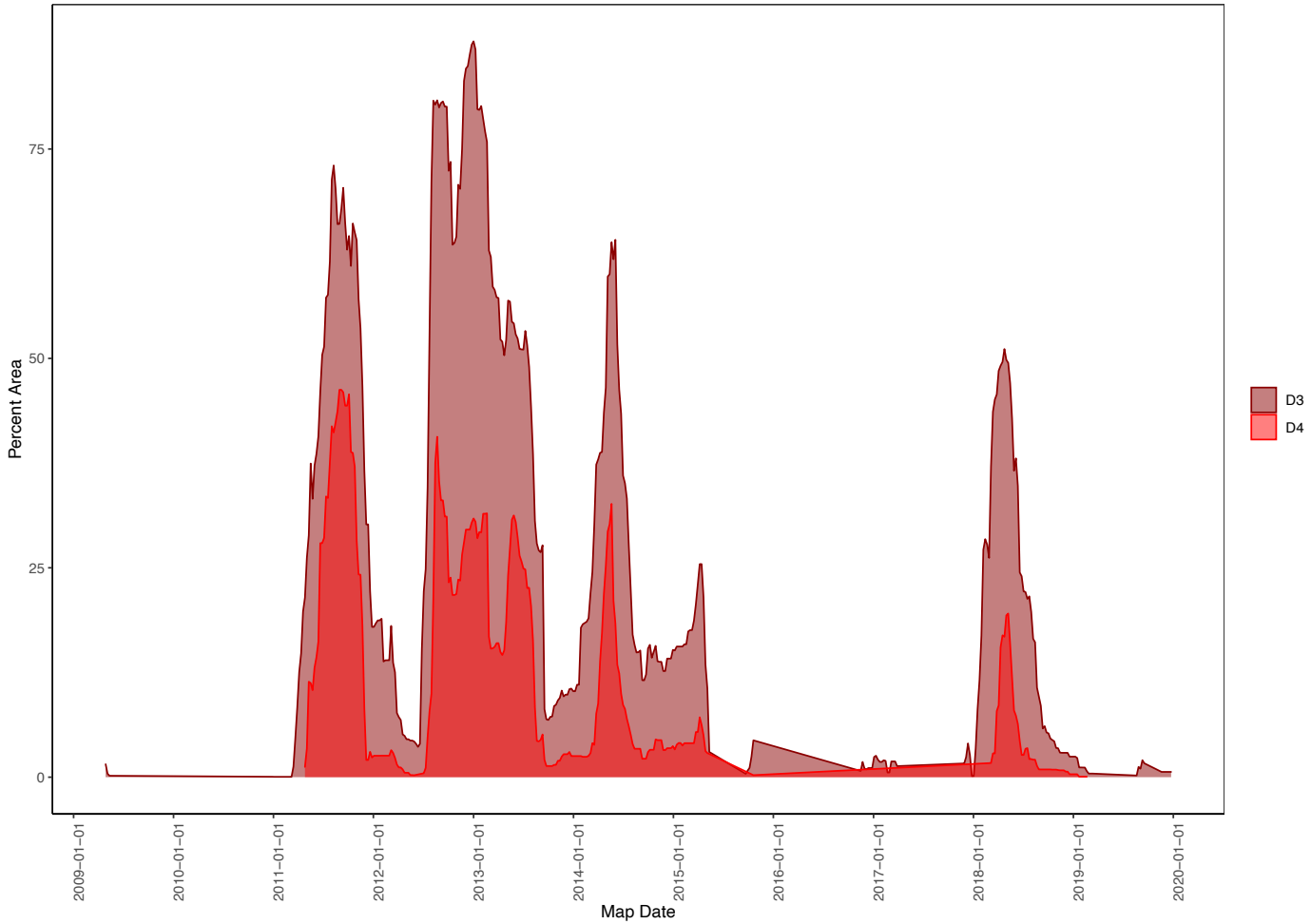

Percent Area
75
50
25
0
2009–01–01
2010–01–01
2011–01–01
2012–01–01
2013–01–01
2014–01–01
2015–01–01
2016–01–01
2017–01–01
2018–01–01
2019–01–01
2020–01–01
Map Date
D3
D4

Table S1 a Pairwise ϕ_{ST} between temporal samples of Arkansas River Shiner collected upstream and downstream of Lake Meredith (LM) below diagonal. **b** Pairwise ϕ_{ST} between temporal samples of Arkansas River Shiner below diagonal (samples upstream and downstream of LM were pooled for 2009 and 2017). P-values are above the diagonal. No values were significantly different from zero.

a

	2009 Upstream LM	**2009 Downstream LM**	**2012 Upstream LM**	**2014 Upstream LM**	**2015 Upstream LM**	**2017 Upstream LM**	**2017 Downstream LM**	**2019 Upstream LM**
2009 Upstream LM	*	0.843	0.268	0.533	0.330	0.576	0.460	0.590
2009 Downstream LM	-0.005	*	0.219	0.735	0.873	0.966	0.405	0.457
2012 Upstream LM	0.002	0.003	*	0.593	0.077	0.235	0.706	0.462
2014 Upstream LM	-0.002	-0.005	-0.003	*	0.377	0.704	0.672	0.797
2015 Upstream LM	0.001	-0.005	0.008	0.0001	*	0.699	0.183	0.070
2017 Upstream LM	-0.002	-0.005	0.002	-0.003	-0.003	*	0.581	0.283
2017 Downstream LM	-0.002	-0.001	-0.007	-0.007	0.007	-0.004	*	0.892
2019 Upstream LM	-0.002	-0.001	-0.0008	-0.003	0.007	0.0007	-0.008	*

b

	2009	**2012**	**2014**	**2015**	**2017**	**2019**
2009	*	0.144	0.551	0.451	0.571	0.359
2012	0.004	*	0.605	0.073	0.312	0.463
2014	-0.002	-0.002	*	0.373	0.759	0.788
2015	-0.0008	0.008	0.0001	*	0.418	0.070
2017	-0.001	0.001	-0.003	-0.001	*	0.403
2019	0.0002	-0.001	-0.003	0.007	0.0003	*

Table S2 Pairwise ϕ_{ST} between temporal samples of Peppered Chub below diagonal. P-values are above the diagonal. There were no significant differences.

	2015	2016	2017	2018
2015	*	0.835	0.988	0.558
2016	-0.005	*	0.356	0.109
2017	-0.007	0.001	*	0.418
2018	-0.003	0.005	-0.0005	*

Table S3 Pairwise ϕ_{ST} between temporal samples of Plains Minnow below diagonal. P-values are above the diagonal. Values were not significantly different from zero.

	2013	2014	2015
2013	*	0.111	0.228
2014	0.012	*	0.197
2015	0.006	0.004	*

Table S4 a Pairwise F_{ST} between temporal samples of Arkansas River Shiner collected upstream and downstream of Lake Meredith (LM) below diagonal. Nine values were significantly different from zero after Bonferroni correction (shaded in grey). **b** Pairwise F_{ST} between temporal samples of Arkansas River Shiner below diagonal. P-values are above the diagonal. Eight values were significantly different from zero after Bonferroni correction (shaded in grey).

a

	2009 Upstream LM	2009 Downstream LM	2012 Upstream LM	2014 Upstream LM	2015 Upstream LM	2017 Upstream LM	2017 Downstream LM	2019 Upstream LM
2009 Upstream LM	*	0.204	0.018	0.000	0.044	0.128	0.179	0.000
2009 Downstream LM	0.001	*	0.085	0.003	0.050	0.105	0.151	0.000
2012 Upstream LM	0.002	0.001	*	0.031	0.056	0.007	0.530	0.001
2014 Upstream LM	0.006	0.004	0.002	*	0.000	0.000	0.051	0.000
2015 Upstream LM	0.002	0.002	0.002	0.007	*	0.019	0.126	0.000
2017 Upstream LM	0.001	0.001	0.002	0.005	0.002	*	0.425	0.000
2017 Downstream LM	0.002	0.002	0.000	0.003	0.002	0.000	*	0.166
2019 Upstream LM	0.006	0.004	0.003	0.005	0.007	0.004	0.002	*

b

	2009	2012	2014	2015	2017	2019
2009	*	0.021	0.004	0.000	0.046	0.000
2012	0.001	*	0.033	0.056	0.016	0.001
2014	0.004	0.002	*	0.000	0.000	0.000
2015	0.002	0.002	0.007	*	0.016	0.000
2017	0.001	0.002	0.004	0.002	*	0.000
2019	0.005	0.003	0.005	0.007	0.004	*

Table S5 Pairwise F_{ST} between temporal samples of Peppered Chub below diagonal. P-values are above the diagonal. A single value was significantly different from zero following Bonferroni correction (2016 to 2017). Peppered Chub are restricted to sites upstream of Lake Meredith.

	2015	2016	2017	2018
2015	*	0.136	0.033	0.382
2016	0.001	*	0.001	0.276
2017	0.002	0.004	*	0.019
2018	0.0002	0.0003	0.002	*

Table S6 Pairwise F_{ST} between temporal samples of Plains Minnow below diagonal. P-values are above the diagonal. The 2013-2014 comparison was significantly different from zero before and after Bonferroni correction. Samples collected above and below Lake Meredith (2013) were pooled because of the small number of samples collected downstream.

	2013	2014	2015
2013	*	0.001	0.034
2014	0.009	*	0.784
2015	0.005	-0.001	*

Literature Cited

Baker DB, Richards RP, Loftus TT, Kramer JW (2004) A new flashiness index: characteristics and applications to midwestern rivers and streams. Journal of the American Water Resources Association 40:503–522.

Dudley RK, Brandenburg WH, Farrington M, Renfro LE, Platania SP (2012) Distribution and conservation status of fishes in the South Canadian River Drainage, New Mexico. Report to New Mexico Department of Game and Fish. https//doi.org/10.13140/RG.2.1.1940.3686